Problem–Solution Fit

Managing employee data manually is time-consuming and often leads to errors, duplication, and difficulty in maintaining records. It also lacks real-time access, proper security, and efficient tracking of attendance, payroll, and leave details.
An automated employee data system addresses these issues by storing all employee information in a centralized database and automating key HR tasks such as attendance tracking, payroll processing, and leave management.

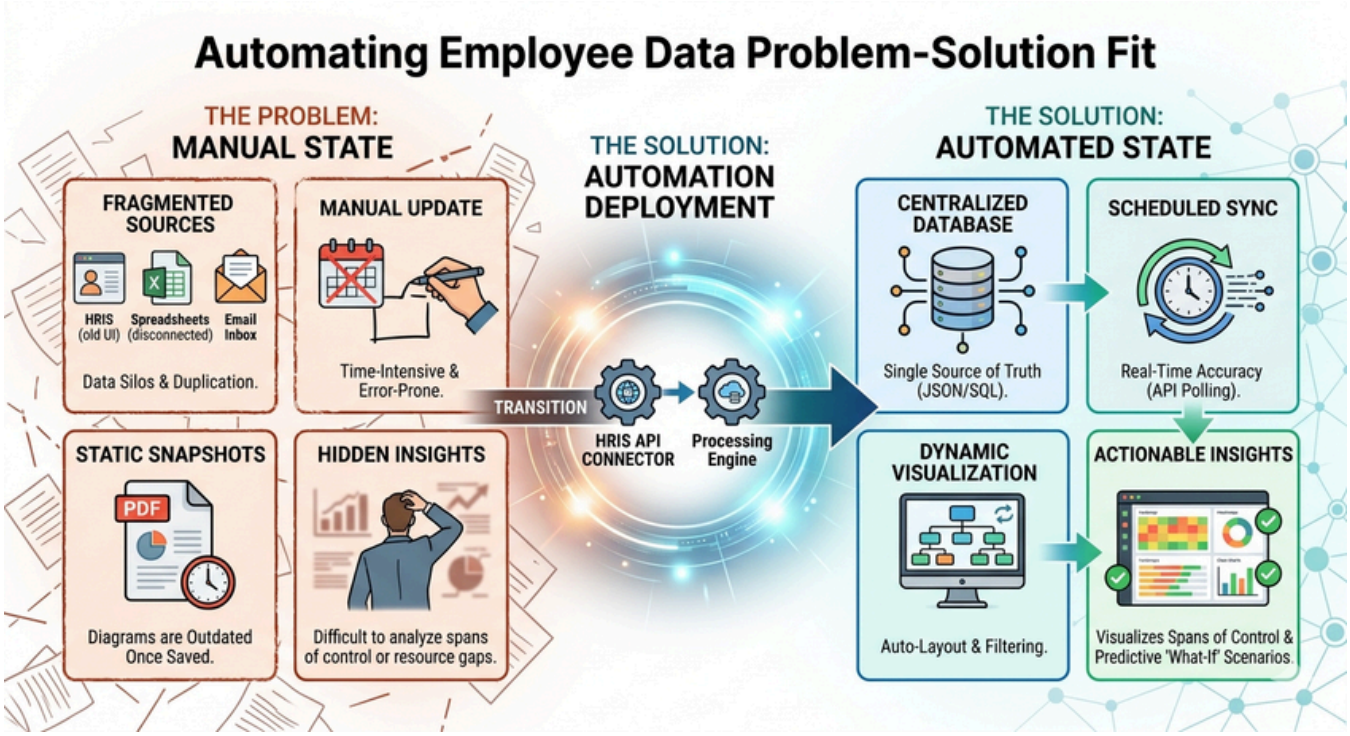
Add a Tools & Technologies
Frontend:

Frontend: HTML, CSS, JavaScript
Backend: Python / Java / Node.js
Database: MySQL / MongoDB
Cloud: AWS / Azure

Purpose:

The purpose of automating employee data is to simplify and improve the management of employee information. It helps reduce manual work, minimize errors, and ensure accurate record-keeping. The system provides quick and real-time access to data, improves efficiency in HR tasks like payroll, attendance, and leave management, and enhances data security. Overall, it saves time, reduces costs, and supports better decision-making in organizations.

Problem solution fit diagram:



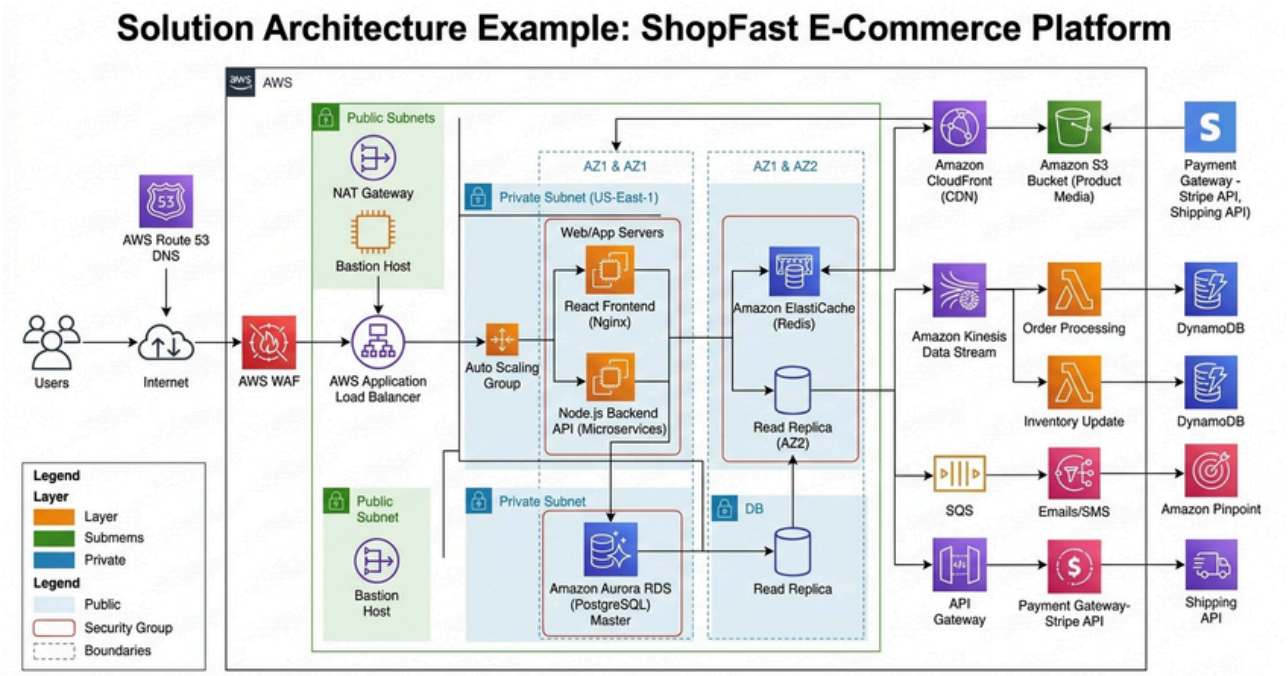
Proposed solution:

Parameter	Description
Data Recency (TTL)	The "Time-to-Live" for data. Most solutions target Real-time or Daily Polls via Webhooks to ensure the diagram reflects today's headcount.
Schema Mapping	The logic that translates HRIS fields (e.g., Reports_To_ID) into visual nodes and edges. It must support recursive hierarchies.
API Throughput	API Throughput The system's ability to handle bulk updates (e.g., during a merger or annual restructuring) without hitting rate limits or crashing the UI.
RBAC Levels	RBAC Levels Role-Based Access Control. Parameters that define who can see sensitive data (Salary, Home Address) versus public data (Name, Title, Dept).

Solution architecture:

Definition:

The architecture for automating employee data visualization moves away from manual data entry and "drawing tools" and instead utilizes an Event-Driven, Service-Oriented Architecture (SOA).

In this model, the "truth" (the core employee data definition) lives solely within the HRIS or Centralized Database. The visualization layer acts as a reactive client, only displaying information that is pushed to it through secured, validated services.

Example: